

INTA 6450

Who Gets Hacked? The Real Drivers of Cyber Conflict

RAJ PATEL

Question

→ Does regime type predict how frequently a country is targeted by state-sponsored cyberattacks? (Option 2)

- Cross-national approach to identify patterns in who gets targeted and why
- We lack systematic understanding of what structural features make a country a recurring target

Contribution

- Prior studies have focused on cyber capabilities, offensive behavior, and attack intent, but not on how often states are targeted or why.
- Prior studies have used shorter time windows

Factors That Motivate
State-Sponsored
Cyberattacks

Lance Y. Hunter, Ph.D.

Craig Douglas Albert, Ph.D.

Eric Garrett

Hypothesis

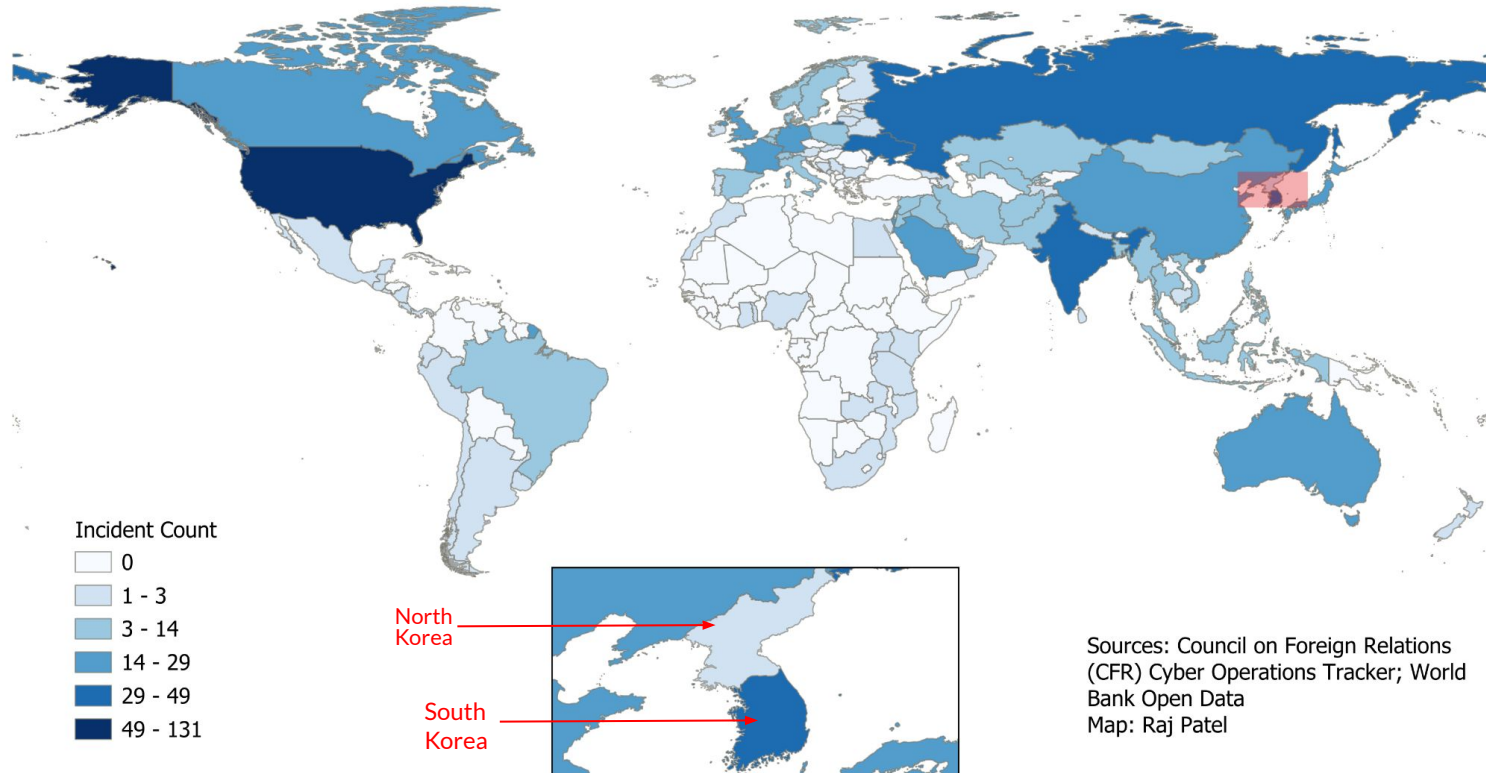
→ **Democracies are targeted by state-sponsored cyberattacks at higher rates than autocracies**

- Open information environments make democracies easier to penetrate
- High-value digital infrastructure makes them more attractive espionage targets
- Democracies just happen to be enemies of countries doing the attacking

Data & Methods

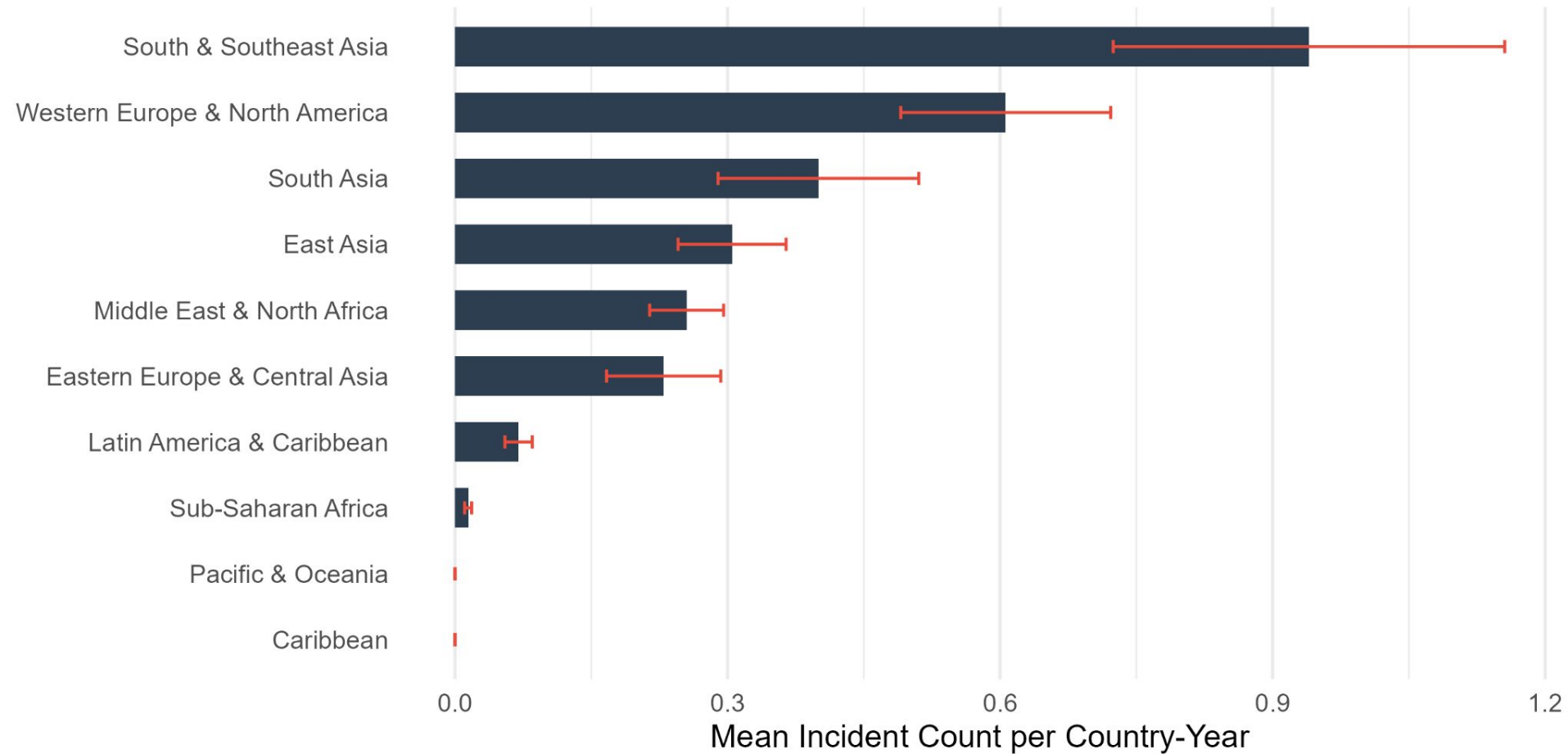
Variable	Operationalization	Data Source	Unit of Analysis
Cyber targeting frequency (DV)	Count of state-sponsored cyber incidents where a country appears as a target	CFR Cyber Operations Tracker	Country-year
Regime type (IV)	VDem Liberal Democracy Index (0–1)	V-Dem dataset	Country-year
Economic development (control)	GDP per capita (constant 2015 USD)	World Bank Open Data	Country-year
Internet penetration (control)	% of population using the internet	World Bank Open Data	Country-year
Govt. Effectiveness (control)	WGI score (-2.5 to +2.5)	World Bank Open Data	Country-year

Global Distribution of State-sponsored Cyberattacks (2005 - 2022)

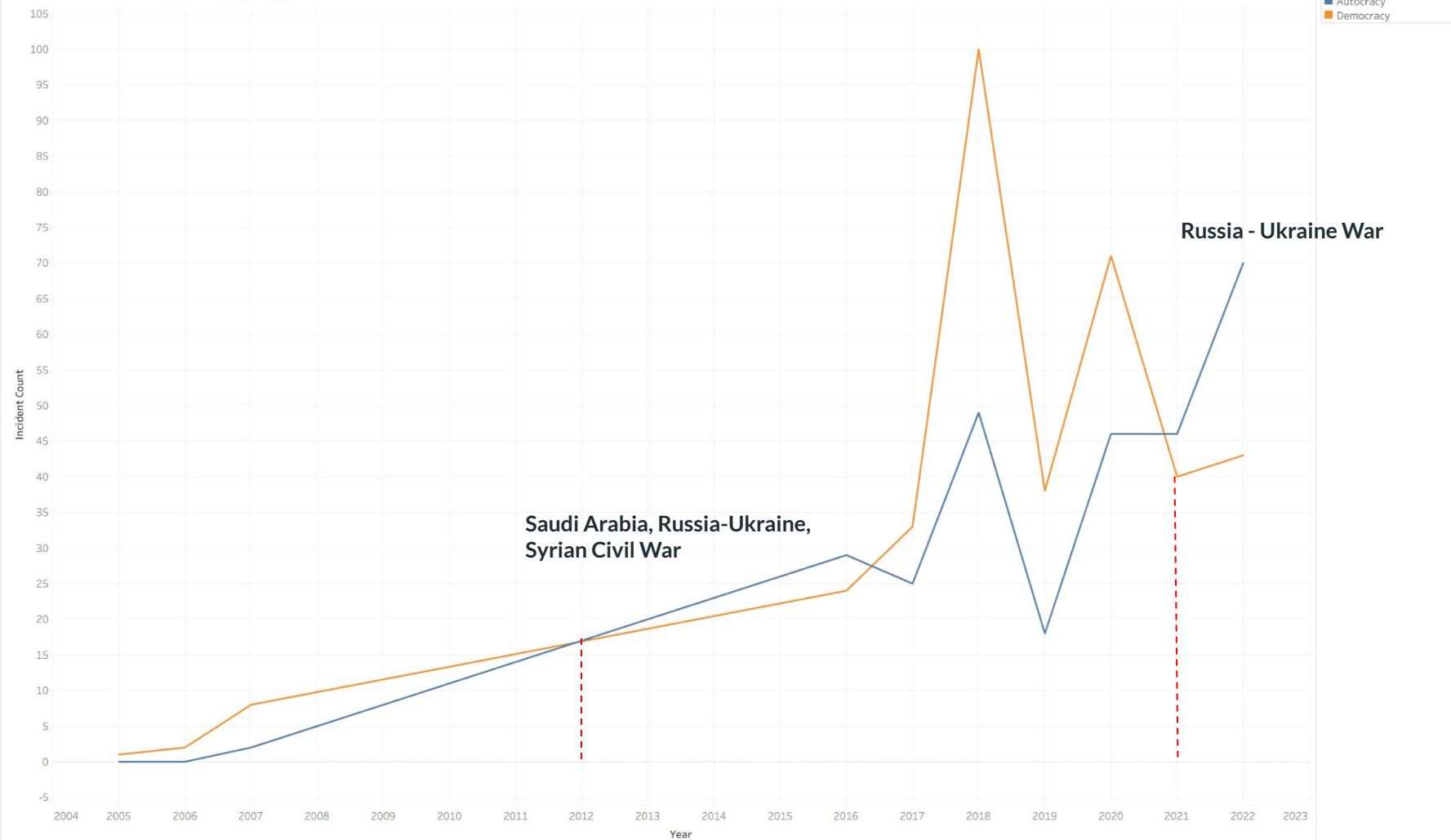


Average State-Sponsored Cyberattack Frequency by Region

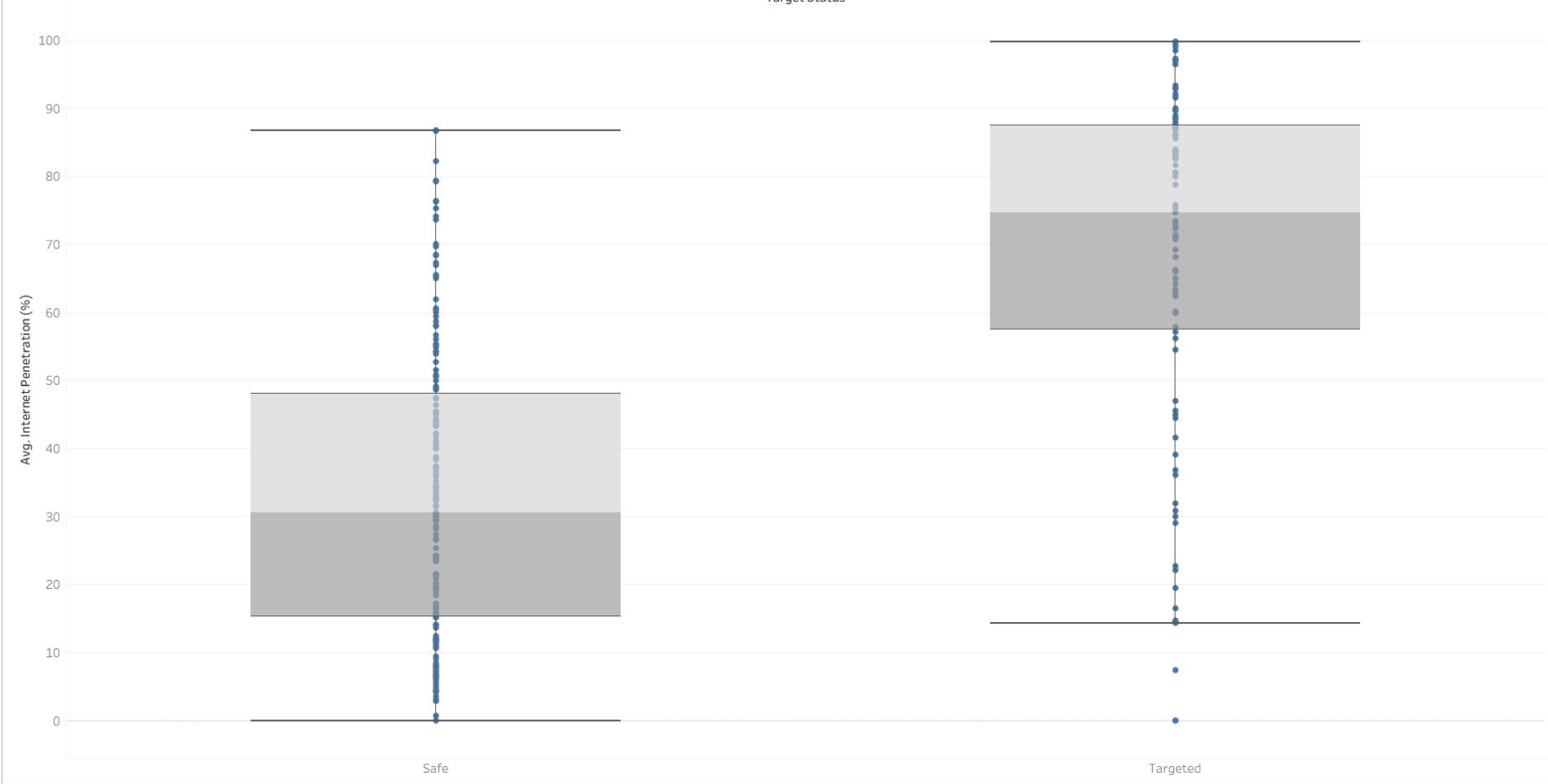
Error bars show ± 1 standard error



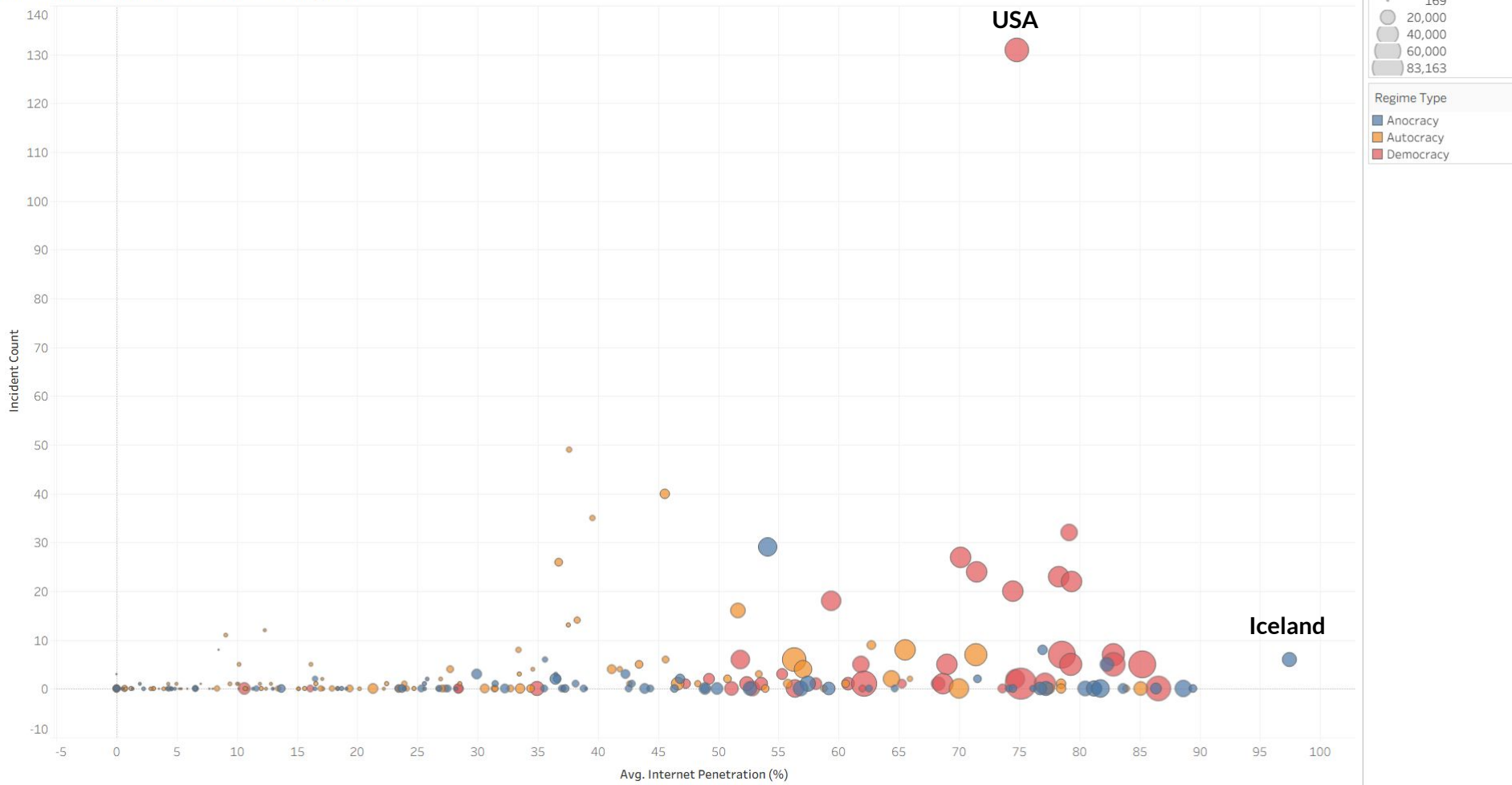
Incidents over Time per Regime Type



Target Status



Internet vs. Attacks vs. GDP vs. Regime



Findings

→ Ran 2 OLS models

→ **1% increase in internet penetration is associated with 0.01 additional attacks per country-year, significant at the 99% level**

Table 1: Predictors of State-Sponsored Cyberattack Frequency

	Dependent variable:	
	Cyberattack Frequency (Count)	
	OLS	
	Base OLS (1)	Full OLS (2)
Liberal Democracy Index (V-Dem)	-0.185 (0.136)	-0.451** (0.203)
Log GDP per Capita	0.018 (0.032)	-0.037 (0.052)
Internet Penetration (%)	0.010*** (0.001)	0.010*** (0.001)
Gov. Effectiveness (Detection Bias)		0.026 (0.074)
Constant	-0.175 (0.212)	0.329 (0.424)
Regional Fixed Effects	No	Yes
Observations	2,482	2,261
R ²	0.056	0.081
Adjusted R ²	0.055	0.076
Residual Std. Error	1.431 (df = 2478)	1.480 (df = 2247)
F Statistic	48.967*** (df = 3; 2478)	15.231*** (df = 13; 2247)

Note:

Robust standard errors recommended. *p<0.1; **p<0.05; ***p<0.01

Conclusion & Future Work

- Internet penetration is the primary structural predictor of cyberattack targeting
- Regime type hypothesis revised - democracy is insignificant in the base model but predicts *fewer* attacks once regional variation is controlled for
- Cyber operations complement conventional conflict
- CFR dataset relies on Western OSINT sources
- Marginal Effects Plot to visualize the predicted probability of attacks based on connectivity and regime type

Questions?